

Simple JWT Auth System

Project Details

Role: Software Developer

Project Type: fullstack

Difficulty: junior

Duration: 3 days

Focus: authentication

Student Name: Rahul Sharma

Description: Build a minimal full stack JWT authentication system using Python and a lightweight frontend.

Technologies

- Python
- FastAPI
- JWT
- SQLite
- HTML/CSS/JS

Requirements

- Create a secure auth flow with JWT.
- Validate user input and handle errors.
- Demonstrate end to end registration, login, and protected access.

Objectives

- Implement secure JWT authentication.
- Validate input and error handling.
- Show full flow from registration to protected dashboard.

Functional Requirements

- Register users with email/password, hash passwords.
- Login returns JWT token.
- JWT required for /dashboard endpoint.
- Validate email format and password length.
- Return appropriate JSON error messages.

Non Functional Requirements

- Store passwords with bcrypt.
- Keep JWT secret in environment variable.
- Use JSON responses with correct status codes.
- Rate limit login attempts to 5/min per IP.

Modules

User Management

Description: Handles user data, hashing, and persistence.

Features

- User model
- Password hashing
- CRUD operations

Auth Service

Description: Provides login, token generation, and middleware.

Features

- Login endpoint
- JWT creation
- Auth middleware

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Status: active